

水溶性维生素

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维生素C

基本特征：

- 1、酸性多羟基化合物， 0.5%的维生素C溶液其 $\text{pH} < 3$ 。
- 2、自然界有两种类型，L和D，L型具有活性。
- 3、结晶维生素C 稳定，但水溶液极易氧化。
- 4、食物中维生素C 有氧化型和还原型，均有生物活性。
- 5、血浆维生素C主要以还原型存在，还原型：氧化型=15:1

吸收和代谢

- 1、维生素C通过扩散或主动运输由肠道进入血液循环。
- 2、维生素C以氧化型被吸收，进入机体后被还原，需谷胱甘肽参与。
- 3、维生素C可在体内有一定量的存储，主要存储在骨骼、脑和肝脏组织。
- 4、维生素C主要通过尿液排出体外，其次为汗液和粪便。

生理功能

- 1、抗氧化作用，清除机体内的氧自由基
- 2、作为羟基过程底物和酶的辅助因子；
- 3、改善铁、钙和叶酸的利用；
- 4、促进类固醇代谢；
- 5、参与合成神经递质，促进大脑合成去甲肾上腺素和5-羟色胺。

缺乏和过量

缺乏：

- 1、牙龈炎
- 2、皮下出血；
- 3、骨质疏松：

过量：

一次大剂量（2-8克）会出现腹泻、腹胀，
患有结石的病人，长期过量摄入会增加尿路结石风险。

机体营养状况评价

血液生化指标：

- 1、尿负荷试验
- 2、血浆维生素C含量测定
- 3、白细胞中维生素C含量测定

参考摄入量及食物来源

推荐摄入量： 14岁及以上人群100mg/天； 预防非传染性慢性疾病的建议量200mg/天。

需要量增加： 高温、寒冷、缺氧的环境、特殊职业人群如接触铅、苯和汞的人群； 孕妇 乳母等

来源

新鲜的蔬菜 水果： 一般叶菜比茎菜多， 酸味水果比无酸味水果含量多。

	维生素C/(mg/d)			
人群	平均需要量EAR	推荐摄入量RNI	建议摄入量PI	可耐受最高摄入量UL
0岁~	—	40 (AI)	—	—
0.5岁~	—	40(AI)	—	—
1岁~	35	40	—	400
4岁~	40	50	—	600
7岁~	55	65	—	1000
11岁~	75	90	—	1400
14岁~	85	100	—	1800
18岁~	85	100	200	2000
50岁~	85	100	200	2000
65岁~	85	100	200	2000
80岁~	85	100	200	2000
孕妇（早）	+0	+0	200	2000
孕妇（中）	+10	+15	200	2000
孕妇（晚）	+10	+15	200	2000
乳母	+40	+50	200	2000

维生素补充剂与健康收益

美国疾病控制和预防中心(CDC)的一项全国健康调查显示：

1988-1994年： 30%的人服用维生素补充剂

2003-2006年： 39%的人服用维生素补充剂

而所有营养剂使用人群从42%上升到了53%。

2010年美国的补充剂市场为280亿美元。

中国维生素和矿物质销售额

2003年： 113.6亿元人民币

2011年： 190亿元人民币

维生素C 能治感冒吗？

维C银翘片为中国国家食品药品监督管理局（食药监局）批准的非处方药（OTC）。

食药监局提供的说明书显示，维C银翘片功能主治为疏风解表，清热解毒。用于外感风热所致的流行性感冒，症见发热、头痛、咳嗽、口干、咽喉疼痛。

每片含维生素C 49.5毫克、对乙酰氨基酚105毫克、马来酸氯苯那敏1.05毫克。

Vitamin C for preventing and treating the common cold.

Douglas RM, Hemilä H, Chalker E, Treacy B.

Update in

Vitamin C for preventing and treating the common cold. [Cochrane Database Syst Rev. 2013]

Abstract

BACKGROUND: The role of vitamin C (ascorbic acid) in the prevention and treatment of the common cold has been a subject of controversy for 60 years, but is widely sold and used as both a preventive and therapeutic agent.

OBJECTIVES: To discover whether oral doses of 0.2 g or more daily of vitamin C reduces the incidence, duration or severity of the common cold when used either as continuous prophylaxis or after the onset of symptoms.

SEARCH STRATEGY: We searched the Cochrane Central Register of Controlled Trials (CENTRAL) (The Cochrane Library Issue 4, 2006); MEDLINE (1966 to December 2006); and EMBASE (1990 to December 2006).

SELECTION CRITERIA: Papers were excluded if a dose less than 0.2 g per day of vitamin C was used, or if there was no placebo comparison.

DATA COLLECTION AND ANALYSIS: Two review authors independently extracted data and assessed trial quality. 'Incidence' of colds during prophylaxis was assessed as the proportion of participants experiencing one or more colds during the study period. 'Duration' was the mean days of illness of cold episodes.

MAIN RESULTS: Thirty trial comparisons involving 11,350 study participants contributed to the meta-analysis on the relative risk (RR) of developing a cold whilst taking prophylactic vitamin C. The pooled RR was 0.96 (95% confidence intervals (CI) 0.92 to 1.00). A subgroup of six trials involving a total of 642 marathon runners, skiers, and soldiers on sub-arctic exercises reported a pooled RR of 0.50 (95% CI 0.38 to 0.66). Thirty comparisons involving 9676 respiratory episodes contributed to a meta-analysis on common cold duration during prophylaxis. A consistent benefit was observed, representing a reduction in cold duration of 8% (95% CI 3% to 13%) for adults and 13.6% (95% CI 5% to 22%) for children. Seven trial comparisons involving 3294 respiratory episodes contributed to the meta-analysis of cold duration during therapy with vitamin C initiated after the onset of symptoms. No significant differences from placebo were seen. Four trial comparisons involving 2753 respiratory episodes contributed to the meta-analysis of cold severity during therapy and no significant differences from placebo were seen.

AUTHORS' CONCLUSIONS: The failure of vitamin C supplementation to reduce the incidence of colds in the normal population indicates that routine mega-dose prophylaxis is not rationally justified for community use. But evidence suggests that it could be justified in people exposed to brief periods of severe physical exercise or cold environments.

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The failure of vitamin C supplementation to reduce the incidence of colds in the normal population indicates that routine mega-dose prophylaxis is not rationally justified for **community use**. But evidence suggests that it could be justified in people exposed to brief periods of severe physical exercise or cold environments.

结论：在普通人群中，维生素C补充剂并不能降低感冒的发生，说明预防性的使用大剂量维生素C并不适合普通人群。

Vitamin C for preventing and treating the common cold.

Hemilä H¹, Chalker E.

⊕ **Author information**

Abstract

BACKGROUND: Vitamin C (ascorbic acid) for preventing and treating the common cold has been a subject of controversy for 70 years.

OBJECTIVES: To find out whether vitamin C reduces the incidence, the duration or severity of the common cold when used either as a continuous regular supplementation every day or as a therapy at the onset of cold symptoms.

SEARCH METHODS: We searched CENTRAL 2012, Issue 11, MEDLINE (1966 to November week 3, 2012), EMBASE (1990 to November 2012), CINAHL (January 2010 to November 2012), LILACS (January 2010 to November 2012) and Web of Science (January 2010 to November 2012). We also searched the U.S. National Institutes of Health trials register and WHO ICTRP on 29 November 2012.

SELECTION CRITERIA: We excluded trials which used less than 0.2 g per day of vitamin C and trials without a placebo comparison. We restricted our review to placebo-controlled trials.

DATA COLLECTION AND ANALYSIS: Two review authors independently extracted data. We assessed 'incidence' of colds during regular supplementation as the proportion of participants experiencing one or more colds during the study period. 'Duration' was the mean number of days of illness of cold episodes.

MAIN RESULTS: Twenty-nine trial comparisons involving 11,306 participants contributed to the meta-analysis on the risk ratio (RR) of developing a cold whilst taking vitamin C regularly over the study period. In the general community trials involving 10,708 participants, the pooled RR was 0.97 (95% confidence interval (CI) 0.94 to 1.00). Five trials involving a total of 598 marathon runners, skiers and soldiers on subarctic exercises yielded a pooled RR of 0.48 (95% CI 0.35 to 0.64). Thirty-one comparisons examined the effect of regular vitamin C on common cold duration (9745 episodes). In adults the duration of colds was reduced by 8% (3% to 12%) and in children by 14% (7% to 21%). In children, 1 to 2 g/day vitamin C shortened colds by 18%. The severity of colds was also reduced by regular vitamin C administration. Seven comparisons examined the effect of therapeutic vitamin C (3249 episodes). No consistent effect of vitamin C was seen on the duration or severity of colds in the therapeutic trials. The majority of included trials were randomised, double-blind trials. The exclusion of trials that were either not randomised or not double-blind had no effect on the conclusions.

AUTHORS' CONCLUSIONS: The failure of vitamin C supplementation to reduce the incidence of colds in the general population indicates that routine vitamin C supplementation is not justified, yet vitamin C may be useful for people exposed to brief periods of severe physical exercise. Regular supplementation trials have shown that vitamin C reduces the duration of colds, but this was not replicated in the few therapeutic trials that have been carried out. Nevertheless, given the consistent effect of vitamin C on the duration and severity of colds in the regular supplementation studies, and the low cost and safety, it may be worthwhile for common cold patients to test on an individual basis.

维生素C的预防性使用可以使成人的感冒时间缩短8%（95%CI 3-12），儿童14%（95%CI 7-21.），维生素C也能缓解感冒的严重程度。但并不影响感冒的发生率。

AUTHORS' CONCLUSIONS:

The failure of vitamin C supplementation to reduce the incidence of colds in the general population indicates that routine vitamin C supplementation is not justified, yet vitamin C may be useful for people exposed to brief periods of severe physical exercise. Regular supplementation trials have shown that vitamin C reduces the duration of colds, but this was not replicated in the few therapeutic trials that have been carried out. Nevertheless, given the consistent effect of vitamin C on the duration and severity of colds in the regular supplementation studies, and the low cost and safety, it may be worthwhile for common cold patients to test on an individual basis whether therapeutic vitamin C is beneficial for them. Further therapeutic RCTs are warranted.

在临床治疗试验中，维生素C对感冒的病程和严重程度并没有影响。

那维生素C 和癌症呢？

有关维生素C 和癌症，这个是目前为止证据最高的研究----干预研究

High-Dose Vitamin C versus Placebo in the Treatment of Patients with Advanced Cancer Who Have Had No Prior Chemotherapy — A Randomized Double-Blind Comparison

Charles G. Moertel, M.D., Thomas R. Fleming, Ph.D., Edward T. Creagan, M.D., Joseph Rubin, M.D., Michael J. O'Connell, M.D., and Matthew M. Ames, Ph.D.

N Engl J Med 1985; 312:137-141 | January 17, 1985 | DOI: 10.1056/NEJM198501173120301

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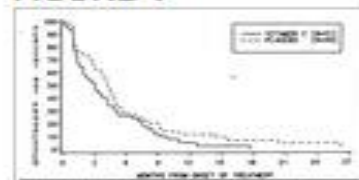
Abstract

It has been claimed that high-dose vitamin C is beneficial in the treatment of patients with advanced cancer, especially patients who have had no prior chemotherapy. In a double-blind study 100 patients with advanced colorectal cancer were randomly assigned to treatment with either high-dose vitamin C (10 g daily) or placebo. Overall, these patients were in very good general condition, with minimal symptoms. None had received any previous treatment with cytotoxic drugs.

Vitamin C therapy showed no advantage over placebo therapy with regard to either the interval between the beginning of treatment and disease progression or patient survival. Among patients with measurable disease, none had objective improvement. On the basis of this and our previous randomized study, it can be concluded that high-dose vitamin C therapy is not effective against advanced malignant disease regardless of whether the patient has had any prior chemotherapy. (N Engl J Med 1985; 312:137-41.)

MEDIA IN THIS ARTICLE

FIGURE 1



Time from the Beginning of Therapy to Disease Progression, According to Treatment Assignment in 100 Patients with Advanced Cancer.

FIGURE 2



Vitamin C therapy showed no advantage over placebo therapy with regard to either the interval between the beginning of treatment and disease progression or patient survival. Among patients with measurable disease, none had objective improvement. On the basis of this and our previous randomized study, it can be concluded that high-dose vitamin C therapy is not effective against advanced malignant disease regardless of whether the patient has had any prior chemotherapy.

高剂量的维生素C补充剂对癌症患者的疾病进展、和预后的作用和安慰剂没有差别。

Lancet. 2004 Oct 2-8;364(9441):1219-28.

Antioxidant supplements for prevention of gastrointestinal cancers: a systematic review and meta-analysis.

Bjelakovic G¹, Nikolova D, Simonetti RG, Gluud C.

⊕ Author information

Abstract

BACKGROUND: Oxidative stress can cause cancer. Our aim was to establish whether antioxidant supplements reduce the incidence of gastrointestinal cancer and mortality.

METHODS: With the Cochrane Collaboration methodology, we reviewed all randomised trials comparing antioxidant supplements with placebo for prevention of gastrointestinal cancers. We searched electronic databases and reference lists (February, 2003). Outcome measures were incidence of gastrointestinal cancers, overall mortality, and adverse effects. Outcomes were analysed with fixed-effect and random-effects model meta-analyses and were reported as relative risk with 95% CIs.

FINDINGS: We identified 14 randomised trials (n=170,525). Trial quality was generally high. Heterogeneity of results was low to moderate. Neither the fixed-effect (relative risk 0.96, 95% CI 0.88-1.04) nor random-effects meta-analyses (0.90, 0.77-1.05) showed significant effects of supplementation with beta-carotene, vitamins A, C, E, and selenium (alone or in combination) versus placebo on oesophageal, gastric, colorectal, pancreatic, and liver cancer incidences. In seven high-quality trials (n=131727), the fixed-effect model showed that antioxidant significantly increased mortality (1.06, 1.02-1.10), unlike the random-effects meta-analysis (1.06, 0.98-1.15). Low-quality trials showed no significant effect of antioxidant supplementation on mortality. The difference between the mortality estimates in high-quality and low-quality trials was significant (Z=2.10, p=0.04 by test of interaction). beta-carotene and vitamin A (1.29, 1.14-1.45) and beta-carotene and vitamin E (1.10, 1.01-1.20) significantly increased mortality, whereas beta-carotene alone only tended to increase mortality (1.05, 0.99-1.11). In four trials (three with unclear or inadequate methodology), selenium showed significant beneficial effect on the incidence of gastrointestinal cancer.

INTERPRETATION: We could not find evidence that antioxidant supplements can prevent gastrointestinal cancers; on the contrary, they seem to increase overall mortality. The potential preventive effect of selenium should be studied in adequate randomised trials.

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Review Antioxidants for prevention of gastrointestinal cancer: a systematic review and meta-analysis. [Cochrane Database of Systematic Reviews 2004, Issue 4.]

Review Systematic review of the effects of antioxidant supplements on the incidence of cancer. [Archives of Internal Medicine 2004; 164(18):1645-1654.]

Review Antioxidants for prevention of gastrointestinal cancer: a systematic review and meta-analysis. [Cochrane Database of Systematic Reviews 2004, Issue 4.]

Do antioxidants prevent gastrointestinal cancer? A meta-analysis. [Gut 2004; 53(12):1715-1721.]

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INTERPRETATION:

We could not find evidence that antioxidant supplements can prevent gastrointestinal cancers; on the contrary, they seem to increase overall mortality.

The potential preventive effect of selenium should be studied in adequate randomised trials.

抗氧化补充剂并不能预防消化道癌症；相反，他们可能增加总死亡风险。

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JAMA. 2005 Mar 16;293(11):1338-47.

Effects of long-term vitamin E supplementation on cardiovascular events and cancer: a randomized controlled trial.

Lonn E¹, Bosch J, Yusuf S, Sheridan P, Pogue J, Arnold JM, Ross C, Arnold A, Sleight P, Probstfield J, Dagenais GR; HOPE and HOPE-TOO Trial Investigators.

[+ Author information](#)

Abstract

CONTEXT: Experimental and epidemiological data suggest that vitamin E supplementation may prevent cancer and cardiovascular events. Clinical trials have generally failed to confirm benefits, possibly due to their relatively short duration.

OBJECTIVE: To evaluate whether long-term supplementation with vitamin E decreases the risk of cancer, cancer death, and major cardiovascular events.

DESIGN, SETTING, AND PATIENTS: A randomized, double-blind, placebo-controlled international trial (the initial Heart Outcomes Prevention Evaluation [HOPE] trial conducted between December 21, 1993, and April 15, 1999) of patients at least 55 years old with vascular disease or diabetes mellitus was extended (HOPE-The Ongoing Outcomes [HOPE-TOO]) between April 16, 1999, and May 26, 2003. Of the initial 267 HOPE centers that had enrolled 9541 patients, 174 centers participated in the HOPE-TOO trial. Of 7030 patients enrolled at these centers, 916 were deceased at the beginning of the extension, 1382 refused participation, 3994 continued to take the study intervention, and 738 agreed to passive follow-up. Median duration of follow-up was 7.0 years.

INTERVENTION: Daily dose of natural source vitamin E (400 IU) or matching placebo.

MAIN OUTCOME MEASURES: Primary outcomes included cancer incidence, cancer deaths, and major cardiovascular events (myocardial infarction, stroke, and cardiovascular death). Secondary outcomes included heart failure, unstable angina, and revascularizations.

RESULTS: Among all HOPE patients, there were no significant differences in the primary analysis: for cancer incidence, there were 552 patients (11.6%) in the vitamin E group vs 586 (12.3%) in the placebo group (relative risk [RR], 0.94; 95% confidence interval [CI], 0.84-1.06; $P = .30$); for cancer deaths, 156 (3.3%) vs 178 (3.7%), respectively (RR, 0.88; 95% CI, 0.71-1.09; $P = .24$); and for major cardiovascular events, 1022 (21.5%) vs 985 (20.6%), respectively (RR, 1.04; 95% CI, 0.96-1.14; $P = .34$). Patients in the vitamin E group had a higher risk of heart failure (RR, 1.13; 95% CI, 1.01-1.26; $P = .03$) and hospitalization for heart failure (RR, 1.21; 95% CI, 1.00-1.47; $P = .045$). Similarly, among patients enrolled at the centers participating in the HOPE-TOO trial, there were no differences in cancer incidence, cancer deaths, and major cardiovascular events, but higher rates of heart failure and hospitalizations for heart failure.

CONCLUSION: In patients with vascular disease or diabetes mellitus, long-term vitamin E supplementation does not prevent cancer or major cardiovascular events and may increase the risk for heart failure.

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In patients with vascular disease or diabetes mellitus, long-term vitamin E supplementation does not prevent cancer or major cardiovascular events and may increase the risk for heart failure.

对于血管疾病或糖尿病患者，长期补充维生素E并癌症或主要的心血管事件的发生没有预防作用，而且增加心衰的风险

Ann Intern Med. 2013 Dec 17;159(12):850-1.

Enough is enough: Stop wasting money on vitamin and mineral supplements.

Guallar E, Stranges S, Mulrow C, Appel LJ, Miller ER 3rd.

Erratum in

Ann Intern Med. 2014 Jan 21;160(2):143.

Comment in

Enough is enough. [Ann Intern Med. 2014]

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Comment on

Long-term multivitamin supplementation and cognitive function in men: a randomized trial. [Ann Intern Med. 2013]

Oral high-dose multivitamins and minerals after myocardial infarction: a randomized trial. [Ann Intern Med. 2013]

Vitamin and mineral supplements in the primary prevention of cardiovascular disease and cancer: An updated systematic evidence review for the U.S. Preventive Services Task Force. [Ann Intern Med. 2013]

PMID: 24490268 [PubMed - indexed for MEDLINE]

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2013年《内科医学年鉴》在回顾了目前有的关于维生素补充剂的大型双盲实验结果后，并认为绝大多数营养补充剂不能预防疾病或者延迟死亡，没有值得信服的医学证据支持普通人群服用维生素补充剂或者矿物质补充剂，尤其是复合维生素。普通人群应该避免服用，因为对他们来说服用的效果弊大于利。